Team 213: New York City Taxi Demand Visualization
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Introduction

NYC taxi drivers rely heavily on personal experience, familiar locations, or visible passenger activity when deciding where to wait or cruise for fares. These decisions require balancing the likelihood of finding a passenger against the time and cost of repositioning, and existing data-driven approaches may still rely heavily on historical patterns or provide limited transparency into why one location is preferable to another.

We propose an interactive visual analytics system that combines historical NYC taxi-trip data with real-time traffic conditions to help drivers compare taxi zones based on demand, profitability, trip efficiency, and congestion. On the computation side, we will use multi-objective Pareto analysis to identify taxi zones that provide favorable tradeoffs among historical demand, profitability, trip efficiency, and current congestion. We may also apply spatial hotspot analysis such as Getis-Ord Gi* to identify statistically significant clusters of high-opportunity areas. On the visualization side, we will use coordinated geographic, temporal, and statistical views so users can interactively explore why different zones are attractive and how those tradeoffs change by time and current traffic conditions.

The project aims to support more informed driver decisions and reduce time spent driving without passengers. Potential secondary benefits include improved fleet planning and broader insights into NYC mobility patterns. We will evaluate the system against established baselines using measures such as pickup success, gross fare and tip revenue per hour, and empty-distance proxies, while considering limitations such as historical data bias, sparse observations in some areas, and interface complexity.

Literature Survey

The literature review done by the group could be summarized using three main themes: analysis of driver behavior and how it translates to profitability, mobility patterns and areas of rider density in NYC, and interactive visual interface design.

For our first theme, Zheng et al. recommended passenger pickup locations by balancing expected profit against travel cost which is relevant to our goal of helping drivers find ideal locations. The shortcoming is its display of information [1]. To improve, we will advance on the interactive visual interface. Building on this, Li et al. uses machine learning to classify driver strategies, hunting vs waiting [2]. This could be helpful in coming up with strategy rules for our project. The shortcoming is evaluating success solely on driving distance. To improve, we will also consider the underlying demand, profitability, and traffic measures. Millard-Ball et al. provides a baseline to compare against as its main idea was around the breakdown of driver actions: cruising, parked, idling, etc. [3]. To improve, we will offer interactive visualization to make more use out of the analysis. Lastly, Qu et al. covers how driver profit is determined by how well drivers spend their travel time rather than just arriving at the closest or most likely pickup location [4]. This supports our reasoning for the project. We will improve by providing a product that drivers can use to take actions that will result in more profitable rides.

Looking at the mobility patterns, Riascos and Mateos, and similarly, Yue et al., clustered pickup and drop off locations to create flow matrices that track passenger movement [5][6]. This is useful to identify dense areas for pickups and dropoffs. The limitation is with the small sample

size and outdated data. To improve, we will look at a larger more recent dataset. Qi et al. incorporates social data, looking at how the number of people getting in and out of taxis changes based on the social function [7]. This is relevant because it can alter predictions in demand. The shortcoming is it divides space into 100m squares. To improve, we can use more realistic neighborhood boundaries. This is similar to what was looked at by Xie et al [8].

Lastly, the visual interface design was by those who explored large-scale NYC taxi trips using spatial, temporal, and origin-destination queries [9]. This is relevant because the system demonstrates how large taxi datasets can be made interpretable through maps, filtering, and coordinated views. However, the visual was designed for urban exploration rather than helping drivers compare profitability or trip efficiency. Coordinated geographic, temporal, and origin-destination visualizations were presented for exploring passenger movement in a large transportation network [10]. Their approach is useful for designing linked maps and time-based views, but it focuses on fixed transit routes rather than taxi-driver economics. We can adapt the coordinated view concept to taxi zones and driver-oriented metrics such as demand, fare, trip duration, and revenue per minute using a mobile app that tells users when taxis are cheaper than Ubers [11]. One of the goals of the paper was to predict demand and show when surge pricing will be activated. We can leverage the visuals of the app to help build our interactive visuals. The shortcomings of this paper is that it uses data from 2014. To improve, we can use more recent data. Chou et al. discusses an interface that forecasts the demand and fare of rides [12]. The paper features Figure 6, a snapshot of their visual interface which can serve as a reference for our end product. To improve, we can update the readability of the visual.

Proposed Method and Implementation Progress

Our methodology is divided into two components: a computational backend that processes large-scale historical taxi data, and an interactive, web-based visual frontend. The innovations for our project will be the calculated metrics listed below. The visualization innovation will be the dynamic pareto overlay.

Our exploratory data analysis follows four stages: data validation, feature construction, spatial-temporal aggregation, and coordinated visual exploration. The objective is to establish reliable inputs for comparing historical taxi activity and revenue-related measures across zones and time periods.

Our preprocessing checks timestamps, zone identifiers, trip distances, durations, and monetary fields. Retained trips must have pickups within their source month, valid pickup and drop-off zones, durations of 1-240 minutes, distances greater than zero and at most 100 miles, and finite, nonnegative required monetary values.

The current pipeline retained **6.77 million August-September 2026 trips**, removing **6.18%** of input records. We use this as a test dataset to develop our algorithms and expand the time range later. We distinguish basic trip validity from metric eligibility: suspicious monetary amounts or implied speeds can exclude a trip from rate calculations while preserving its contribution to recorded pickup counts. This prevents anomalous values from distorting financial comparisons without unnecessarily discarding activity observations.

We derive occupied duration, pickup date, weekday, and time window, then aggregate by pickup zone. Aggregates preserve counts, monetary sums, occupied-minute totals, and metric-specific denominators. This allows the backend to combine time windows consistently rather than averaging previously calculated rates.

We calculate historical metrics for each selected pickup zone and time period. Where applicable, formulas used to calculate each metric are available in Appendix B.

- **Pickup activity**: retained pickup count divided by the number of selected calendar days, allowing comparisons across periods with different exposure.
- **Fare-plus-tip rate**: summed eligible base fares and recorded tips divided by the corresponding summed occupied minutes. This measures revenue-related activity during passenger trips; it excludes operating costs and empty driving time.
- **Average occupied trip duration**: total occupied minutes divided by the number of selected trips.
- **Credit-card tipping propensity**: eligible credit-card trips with recorded tips exceeding 20% of base fare divided by all eligible credit-card trips with positive base fare. Cash transactions are excluded because cash tips are unrecorded.
- **Origin–destination probability**: trips from zone A to zone B divided by all selected trips departing zone A.
- **Pareto tradeoff analysis:** We identify non-dominated zones using recorded pickups per day and fare-plus-recorded-tip per occupied minute. Zones on the Pareto frontier represent alternative tradeoffs between historical activity and receipt rates. Comparisons use a common time selection and minimum-sample requirements.

Visualization and backend integration

We are developing a custom web dashboard that connects geographic patterns with statistical comparisons to support exploration of historical taxi activity.

Geographic Views: The main part of the dashboard is an interactive choropleth map built with *D3*. We will map the taxi zones and use custom color scales to fill the geographic areas based on their computed attractiveness and revenue hotspots. The dashboard provides date, weekday, and time-window filters for comparing historical taxi activity across zones. Statistical views display recorded pickups per day, fare-plus-recorded-tip per occupied minute, and average occupied trip duration. An interactive tradeoff chart compares pickup activity with fare-plus-tip rates and highlights Pareto-efficient zones.

Filtered Statistical Views: To visualize how revenue opportunities change by time and traffic, we will integrate *Vega-Lite* to render charts. These views will allow users to apply custom time filters and interactively explore underlying driver-oriented metrics like average trip duration.

Observable Plot might be added for a faster, lighter grammar-of-graphics engine that overlays dynamic Pareto frontiers onto your scatter plots.

Backend Integration: The dashboard retrieves filtered results from a Python backend using precomputed counts, monetary sums, occupied-minute totals, and metric-specific denominators. When users change the selected period, the backend combines these quantities and recalculates rates, ensuring consistent aggregation across time windows.

Preliminary Results and Planned Evaluation

To objectively evaluate the effectiveness of our visualization system, we aimed to have an experiment utilizing a historical holdout testing method. We will train our Pareto and Gi* hotspot algorithms on taxi data from 2018–2019. We will use a holdout set (e.g., a specific week in 2020

or 2021) as our "live" testing environment. Then, we will simulate driver shifts and measure the outcomes using the following metrics:

- **Gross Revenue per Hour:** Calculating the combined gross fare and tip revenue generated per hour of driving.
- **Pickup Success & Trip Efficiency:** Measuring the time elapsed between drop-offs and subsequent pickups.

Also, during these experiments, we will test the system's robustness against limited dataset areas, specifically evaluating how the algorithms perform in areas with sparse observations or when historical data is limited.

Project Plan, Budget, & Milestones

This project uses publicly available NYC taxi trip data with no cost as the main resource requirement is computational capacity to store, clean, and analyze the large dataset. The midterm milestone will deliver a minimum viable product connecting the dashboard to a representative sample of NYC taxi dataset zones. This version will feature a basic zone map displaying historical pickup demand, average trip duration, gross receipts, and key revenue hotspots across custom time filters. For the final milestone, the fully deployed application will run on the complete dataset with
smooth filtering and interactive map exploration. Users will be able to evaluate historical opportunities by integrating demand, earnings proxies, and trip efficiency metrics simultaneously. Finally, the completed tool will undergo peer usability testing to validate that its navigation, visual maps, and key metrics are clear and intuitive. The Gantt chart pictured is the same from the project proposal. We are on track and all milestones remain the same. **All team members have contributed a similar amount of effort as seen in the Gantt chart.**

Appendix A.

Literature Summaries

“Mining open datasets for transparency in taxi transport in metropolitan environments” covers the development of a mobile application intended to help riders compare rates between traditional taxis and Uber (Noulas et al. 1). One of the goals of the paper was to predict demand and show when surge pricing will be activated. At those locations and times will be when demand is highest which is what we are looking for. The shortcomings of this paper is that it uses data from 2014. To improve, we can use more recent data.

“Taxi Demand and Fare Prediction with Hybrid Models” discusses the creation of an interface that forecasts the demand and fare of rides. One of the most interesting aspects of this paper is its Figure 6, a snapshot of their visual interface (Chou et al. 12). We would like to use it as a reference for our project. A shortcoming of the visual is the units used. Since the population in NYC would likely use imperial units, our report will improve upon what users will better understand.

Li et al. (2011) dug into a massive taxi dataset to see whether drivers do better by "hunting" (driving around looking for fares) or "waiting" (staying parked at a hotspot). They used a ML model to classify top-performing drivers based on these strategies. It proves we can advise drivers on the best cruising behavior for specific hours like showing that hunting works better during the morning commute, but waiting is better during midday traffic. The main shortcoming

is that they only looked at occupied driving distance to measure success, ignoring actual money made which we are addressing that by using actual NYC data on fares, tolls, and tips to measure profitability.

Yue et al. (2009) tried to find "attractive areas" by clustering pickup and drop-off points, and then built a flow matrix to see how passengers move between these clusters. This flow concept is to help drivers figure out where to head next after they drop someone off. But their experiment was pretty limited, tracking only 480 cars on a single Sunday, which is way too small sample size, which NYC taxi dataset would be much larger.

Another aspect from Qi et al. (2011) is incorporating other social data. They looked at how the number of people getting in and out of taxis changes based on the social function like whether an area is a commercial center or an entertainment district. They found that different types of neighborhoods have totally different demand rhythms, which provide fundamental logic we need to build our temporal filters. However, they divided the city up into rigid 100-meter squares, which might not be how actual NYC neighborhoods and streets work.

Spatiotemporal travel demand[9] used network theory techniques to reveal some correlation in the NYC community. It contains spatiotemporal phenomenon rather than counting citywide trips where it shows hotspot patterns. They found strong spatial clustering across peak periods and showed that pickup/destination patterns differ meaningfully between weekday commutes and weekend leisure periods. It also shows that hotspot patterns reflect real temporal context: commute, commercial activity, transit hubs, entertainment, and weekend travel.

Operational problems[5] were discussed where drivers decide to park, cruise, or reposition while seeking the next fare. The study found that repositioning represented 63% of time between trips, 23% in cruising and 14% in parking. It also notes that taxi cruising and repositioning in NYC historically account for 44% of taxi miles, with about 2.9 deadhead miles between trips.

A driver's profit[4] is determined by how well they spend their travel time rather than just arriving at the closest or most likely pickup location. Using past taxi GPS traces, it finds lucrative routes and suggests a net-profit goal for route ranking.

The paper from Riascos and Mateos maps taxi pickup and drop off locations to create a network that connects the different high demand areas in NYC (1). Connecting popular routes can show where drivers will likely end up after picking up customers from a demand area. Knowing this can help drivers go from one demand area to another instead of selecting a customer who needs to be dropped off at a location where the likelihood of a new customer is low. The shortcoming of this paper is outdated data from 2009-2015.

In “Visual Exploration of Big Spatio-Temporal Urban Data”, Ferreira et al. introduced TaxiVis, an interactive system for exploring large-scale NYC taxi trips using spatial, temporal, and origin-destination queries [2]. The system demonstrates how large taxi datasets can be made interpretable through maps, filtering, and coordinated views. However, TaxiVis is primarily designed for urban exploration rather than specifically helping drivers compare profitability or trip efficiency.

In “Where to Find My Next Passenger?”, Zheng et al. use historical taxi GPS trajectories to recommend locations where drivers are more likely to find passengers while balancing expected profit against travel cost [11]. This is directly relevant to our goal of helping drivers compare attractive locations, but the system emphasizes on a machine learning recommendation system.

Our project instead exposes the underlying demand, profitability, and traffic measures through an interactive visual interface.

In “Visualizing Mobility of Public Transportation System” by Zeng et al. present coordinated geographic, temporal, and origin-destination visualizations for exploring passenger movement in a large transportation network [13]. Their approach is particularly useful for designing our linked map and time-based views, but it focuses on fixed transit routes rather than taxi-driver economics. We can adapt the coordinated view concept to taxi zones and driver-oriented metrics such as demand, fare, trip duration, and revenue per minute.

Appendix B.

Pickup activity

$$Pickup\ Activity\ =\ \frac{Pickup\ Count}{Number\ of\ Days\ in\ Timeframe}$$

Fare-plus-tip rate

$$Fare\ plus\ tip\ rate\ =\ \frac{\Sigma(Base\ fare + Tip)}{\Sigma(trip\ duration\ minutes)}$$

Average occupied trip duration

$$Average\ Trip\ Duration\ =\ \frac{\Sigma(Occupied\ Minutes)}{DistinctCount(Number\ of\ Trips)}$$

Credit-card tipping propensity

$$Tipping\ Propensity\ =\ \frac{Number\ of\ trips\ paid\ with\ credit\ cards\ with\ tip > 20\%\ of\ base\ fare}{DistinctCount(Number\ of\ Trips\ paid\ with\ credit\ card)}$$

Origin–destination probability

$$P(A \rightarrow B)\ =\ \frac{Number\ of\ Trips\ from\ A\ to\ B}{Number\ of\ trips\ with\ pickup\ zone\ A}$$